



January 2025

Failed Zero Forcing Numbers of Trees and Circulant Graphs

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Recommended Citation

Gomez, Luis; Rubi, Karla; Terrazas, Jorden; Florez, Rigoberto; and Narayan, Darren A. (2025) "Failed Zero Forcing Numbers of Trees and Circulant Graphs," *Theory and Applications of Graphs*: Vol. 11: Iss. 1, Article 5.

DOI: 10.20429/tag.2024.110105

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Failed Zero Forcing Numbers of Trees and Circulant Graphs

Cover Page Footnote

We are immensely grateful to an anonymous referee for their suggestions (including a shorter proof for Theorem 5 which helped improve the presentation of our paper. This research was done as part of the National Science Foundation Research Experiences for Undergraduates Program in Extremal Graph Theory and Dynamical Systems (Award No. 1950189) at the Rochester Institute of Technology.

Abstract

Given a graph G , the zero forcing number of G , $Z(G)$, is the smallest cardinality of any set S of vertices on which repeated applications of the forcing rule (described below) results in all vertices being in S . The forcing rule is as follows: if a vertex v is in S , and exactly one neighbor u of v is not in S , then u is added to S in the next iteration. Zero forcing numbers have attracted great interest over the past 15 years and have been well studied. The zero forcing number is used to study the maximum nullity/minimum rank of the family of symmetric matrices described by G . In this paper we investigate the largest size of a set S that does not force all of the vertices in a graph to be in S . This property first studied by Fetcie, Jacob, and Saavedra, is known as the failed zero forcing number of a graph and will be denoted by $F(G)$. This property has been investigated in recent years, including a proof by Shitov that the determination of $F(G)$ is NP-complete. In this paper we present new results involving $F(G)$ for trees including a lower bound based on the independence number. In addition we develop new extremal methods and use them to determine $F(G)$ for families of circulant graphs.

Keywords: zero forcing; failed zero forcing; trees; circulant graphs

MSC Classification: 05C15 (05C50 05C57 05C78)

1 Introduction

Given a graph G , the zero forcing number of G , $Z(G)$, is the smallest cardinality of any set S of vertices on which repeated applications of the forcing rule results in all vertices being in S . The forcing rule is: if a vertex v is in S , and exactly one neighbor u of v is not in S , then u is added to S in the next iteration. The zero forcing number is used to study the maximum nullity/minimum rank of the family of symmetric matrices described by G [2]. Zero forcing numbers have attracted great interest over the past 15 years and have been well studied [5].

In this paper we investigate the largest size of a set S that does not force all of the vertices in a graph to be in S . This quantity, known as the failed zero forcing number of a graph will be denoted by $F(G)$, and was first investigated by Fetcie, Jacob, and Saavedra [6]. A basic result from their paper showed that for any graph G with n vertices, $0 \leq F(G) \leq n - 2$, and these bounds are tight. It was shown by Shitov [10] that the determination of $F(G)$ in general is NP-complete. Fetcie, Jacob, and Saavedra characterized all graphs with a failed zero forcing number of 0 or 1 and posed the problem of characterizing all graphs with a failed zero forcing number of 2 [6]. This was recently solved by Gomez, Rubi, Terrazas, and Narayan [7] where they identified the 15 graphs with a failed zero forcing number of 2. A streamlined proof was given by Kaudan, Taylor, and Narayan [9], who also considered the problem of realizing which graphs have a given subset of vertices as a failed zero forcing set. Recently, Swanson and Ufferman [11] answered a question of Fetcie, Jacob, and Saavedra [6] if for a given k there exists an l such that all graphs with at least l vertices satisfy $F(G) \geq k$. They showed that for a graph with n vertices, $F(G) \geq \lfloor \frac{n-1}{2} \rfloor$.

Independently, a closely related property called the zero blocking number of a graph was introduced by Beaudouin-Lafona, Crawford, Chen, Karst, Nielsen, and Sakai Troxell [3] and Karst, Shen, and Vu [8]. The zero blocking number of a graph G equals $|V(G)| - F(G)$. A variation of the failed zero forcing number, known as the failed skew forcing number was studied by Ansill, Jacob, Penzellna, and Saavedra [1].

We will consider vertices in the set S to be filled and the vertices not in S to be unfilled. When applying the color change rule, we will say a unfilled vertex is forced if it is changed to filled. If a zero forcing set is present we will say that all vertices in the graph are forced or simply the graph is forced.

We will use P_n to denote a *path* with n vertices. A *subdivided star* is a star graph $K_{1,n-1}$ obtained by a sequence of edge-subdivisions where edge(s) $u - v$ are replaced with path(s) $u - w - v$ that includes a new vertex w . The *distance* between two vertices u and v is the maximum number of edges in a shortest $u - v$ path. The *eccentricity* of a vertex w is the maximum distance between w and all vertices in the graph. The

center of a graph is a vertex with minimum eccentricity. The independence number of a graph G , denoted by $\alpha(G)$, is the size of a largest subset of vertices such that no two vertices are adjacent.

We investigate extremal properties of failed zero forcing sets which lead to both new lower bounds and upper bounds. The lower bounds are obtained by construction, and for the upper bounds we show that exceeding a bound means that there are too many filled vertices that are in close proximity of one another, which results in all of the vertices in the graph being forced. In Section 2 we show a connections between the failed zero forcing number and the independence number. We build on the observation by Shitov [10] that in terms of complexity the problem of determining $F(G)$ can be reduced to determining the independence number. Although these properties are closely related, a key difference is that a vertex v of degree 1 can be in an independent set, but v cannot be in a failed zero forcing set unless its neighbor is as well. However we present a surprising result that for trees (including those with many vertices of degree 1), we maintain the bound $F(G) \geq \alpha(G) - 1$.

In Section 3 we determine the failed zero forcing number for certain families of circulant graphs. The lower bounds are obtained by constructing a base case and then using a particular subgraph to extend the graph and a corresponding failed zero forcing set. For the upper bounds we use the pigeonhole principle to show that if there are too many filled vertices close together then all of the vertices in the graph will be forced. We first investigate circulant graphs that are 4-regular and then look at the more complex family of circulant graphs of larger degree.

2 Failed zero forcing numbers and independence numbers

Motivated by a result by Shitov [10] which ties the the complexity of finding the failed zero forcing number to the determination of the independence number, we further investigate the relationship between these two properties. In this section, we show that for trees, and graphs where every vertex has degree at least two, the failed zero forcing number of a graph can be bounded from below by one less than the independence number. In Theorem 2 we present a result involving subdivided stars, which will lead to a more general result involving trees.

2.1 Failed zero forcing numbers of sub-divided stars

We begin by restating a result of Fetcie, Jacob, and Saavedra [6].

Lemma 1. [6] The failed zero forcing number of a path P_n on n vertices is $F(P_n) = \lceil \frac{n-2}{2} \rceil$.

Proof. If S is a failed zero forcing set in P_n then neither end vertex is in S because either end vertex is a zero forcing set. Furthermore, S contains no pairs of adjacent vertices because any pair of adjacent vertices is a zero forcing set. Therefore, S can have at most $\lceil \frac{n-2}{2} \rceil$ vertices. We next show that this upper bound can be achieved. Clearly if a vertex of degree 1 on a path is filled, then the entire path is forced. Let the vertices of P_n be $v_1, v_2, \dots, v_n$. If n is odd, then vertices $v_2, v_4, \dots, v_{n-1}$ form a failed zero forcing set with $\frac{n-1}{2} = \lceil \frac{n-2}{2} \rceil$ vertices. If n is even, then vertices $v_2, v_4, \dots, v_{n-2}$ form a failed zero forcing set with $\frac{n-2}{2} = \lceil \frac{n-2}{2} \rceil$ vertices. $\square$

We use the above result in the determination of the failed zero forcing numbers of subdivided stars.

Theorem 2. Assume G is a subdivided star graph with the center vertex being v_1 . Let the paths incident to v_1 be $P_{\alpha_1}, P_{\alpha_2}, \dots, P_{\alpha_t}$. Then $F(G) = \max_{\alpha_i \neq \alpha_j} \{F(P_{\alpha_i}) + F(P_{\alpha_j}) + 1 + \sum_{k \neq \alpha_i, \alpha_j} |V(P_k)|\}$.

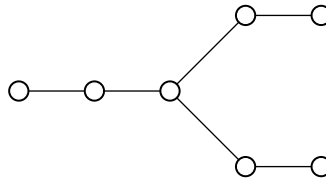


Figure 1: A subdivided-star graph

Proof. Let the paths incident to the center vertex v_1 , be $P_{\alpha_1}, P_{\alpha_2}, \dots, P_{\alpha_t}$. We note that if there are two adjacent vertices that are filled, all of the vertices on the entire path are forced. Hence it is impossible to have all but one path with adjacent filled vertices. Hence we have an upper bound where all of the vertices in all but two paths are filled and the two remaining paths do not have adjacent filled vertices. Hence,

$$F(G) \leq \max_{\alpha_i \neq \alpha_j} (F(P_{\alpha_i}) + F(P_{\alpha_j}) + 1 + \sum_{k \neq \alpha_i, \alpha_j} |V(P_k)|)$$

Since all of the paths meet at v_1 , we can fill in v_1 and all vertices in all but two paths to create a failed zero forcing set. In order to get a maximum sized failed zero forcing set however, we could fill in vertices of the two unfilled paths using Lemma 1. This will leave the two vertices adjacent to the central vertex unfilled thus stalling v_1 and still filling in the maximum amount of vertices. We can now sum the vertices in all paths except two along with the failed-zero forcing numbers of P_{α_i} and P_{α_j} and v_1 to obtain the failed-zero forcing number of the entire graph. Combining this with our upper bound gives the intended result.

$$F(G) = \max_{\alpha_i \neq \alpha_j} (F(P_{\alpha_i}) + F(P_{\alpha_j}) + 1 + \sum_{k \neq \alpha_i, \alpha_j} |V(P_k)|)$$

□

2.2 Connections between $F(G)$ and $\alpha(G)$

In this section we will show that for some graph families, exact values or lower bounds for the failed zero forcing number $F(G)$ can be obtained using the independence number of a graph $\alpha(G)$. In Theorem 3, we show that when $\delta(G) \geq 2$, $F(G) \geq \alpha(G)$. However, a key difference between independent sets and failed zero forcing sets is a vertex v of degree 1 can be in an independent set but v can only be in a failed zero forcing set S if its neighbor is also in S .

We next show that the straightforward result that if a graph does not have any vertices of degree 1 then the failed zero forcing number is greater than or equal to the independence number. For the sake of completeness, we include the details.

Theorem 3. *If G is a graph where $\delta(G) \geq 2$, then $F(G) \geq \alpha(G)$.*

Proof. Since $\delta(G) \geq 2$, vertices in an independent set of vertices A will be adjacent to at least two vertices that are not in A . Hence, if all of the vertices in A are filled we have a failed zero forcing set of size $\alpha(G)$. □

It was shown by Fetcie, Jacob, and Saavedra [6] that the failed zero forcing number of a complete graph on n vertices, K_n , is $n - 2$. Since $\alpha(K_n) = 1$ we have examples of strict inequality. We note that the $\delta \geq 2$ property is needed, since for all $k \geq 1$, $F(P_{2k+1}) = k$ but $\alpha(P_{2k+1}) = k + 1$.

Theorem 4. *If G is a cycle, then $F(G) = \alpha(G)$.*

Proof. This follows since S is a failed zero forcing set if and only if S is an independent set. If S is an independent set then each vertex in S is adjacent to two vertices not in S , so S must be a failed zero forcing set. To see the other direction note that if S is not an independent set where $S \subset V(G)$ then there is a vertex $u \in S$ that is adjacent to only one vertex in S . Then u will force its other neighbor and after iterating this process S will not be a failed zero forcing set. By taking a largest independent set we have a failed zero forcing set of maximum size. $\square$

Next we will prove that for any tree T , the independence number cannot be more than 1 greater than the failed zero forcing number.

Theorem 5. *For any tree T , $F(T) \geq \alpha(T) - 1$.*

Proof. For a tree with duplicated leaves, we can color all of the vertices except for the two leaves and obtain $F(T_n) \geq n - 2$, which exceeds $\alpha(T) - 1$. We proceed under the assumption that T does not have any duplicated leaves. Any tree T must contain a vertex v with degree at least 3, where $T - v$ contains two disjoint paths P_1 and P_2 . Let $P = P_1 \cup P_2 \cup \{v\}$ and $V' = V(T) - P$. Let A be an independent set in T by filling all of the vertices in $V' \cup \{v\}$ and as many vertices in P we can obtain a failed zero forcing set with at least $|A \cap P| - 2$ filled vertices in P and at least $|A \cap V'| + 1$ filled vertices in V' , which gives a total of at least $\alpha(T) - 1$ vertices. $\square$

Next we investigate failed zero forcing numbers of trees. Fetcie, Jacob, and Saavedra [6] defined a module of order X to be a set of X vertices that have the same set of neighbors in $V - X$. We restate a result of their paper.

Lemma 6. [6] For a graph G , $F(G) = n - 2$ if and only if G has a module of order 2.

Using a result of Erdős and Rényi we can state the following result.

Theorem 7. *Almost all trees on n vertices have a failed zero forcing number of $n - 2$.*

Proof. Consider a randomly selected tree T_n . By Theorem 6 in [4] the probability that this tree contains at least one pendant $K_{1,2}$ (called a ‘cherry’) is 1. Since this cherry forms a module, $F(T_n) = n - 2$. $\square$

3 Circulant Graphs

The circulant graph $C_n\{A\}$, where $\{A\} \subset [n]$, is the graph with set of n vertices $v_1, v_2, \dots, v_n$ and set of edges equal to $\{v_i v_j : (i - j) \bmod n \in A \text{ or } (j - i) \bmod n \in A\}$. In all cases, we will assume that $1 \in A$. We will refer to the cycle $v_1, v_2, \dots, v_n$ and edges $v_1 v_2, v_2 v_3, \dots, v_{n-1} v_n, v_n v_1$ as the ‘main cycle’ of the circulant graph.

In this section we will investigate failed zero forcing numbers of circulant graphs. These graphs are appealing to study because of their rotational symmetry. We present numerous results for circulant graphs that are 4-regular and also explore the general case of $2k$ -circulant graphs. We will refer to a set of consecutively filled vertices as a *string* and a set of strings with $k_1, k_2, \dots, k_t$ vertices separated by an unfilled vertex as a $k_1 - k_2 - \dots - k_t$ as a *configuration*. The set of vertices between two unfilled vertices will be known as a *subdivision*. We will analyze which configurations of sets of filled vertices force the graph and which are failed zero forcing sets.

Definition. A configuration $k_1 - k_2 - \dots - k_t$ is a path of sets of $k_i, 1 \leq i \leq t$ consecutively filled vertices where sets are separated by an unfilled vertex between them. Unless stated otherwise the vertex on each end of a configuration is assumed to be unfilled.

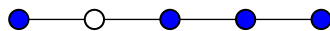
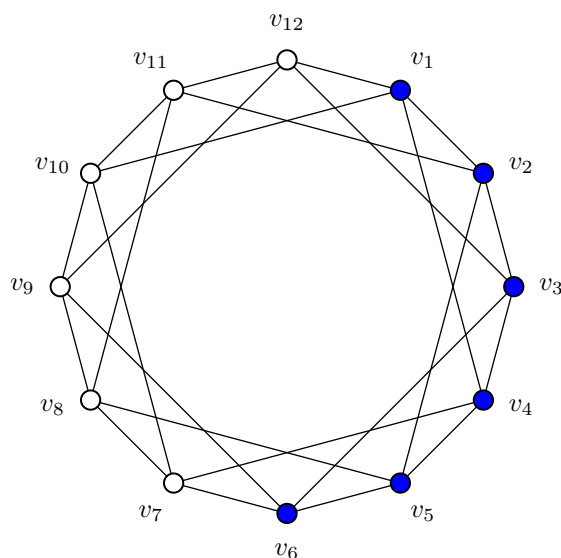


Figure 2: A 1 – 3 configuration

An example of a 1 – 3 configuration is shown in Figure 2.

We next investigate the minimum number of consecutive filled vertices along the main cycle that are needed to force all of the remaining vertices in the graph.

Example. We show that a string of 6 consecutively filled vertices is a forcing set of $C_{12}\{1, 3\}$. Let the vertices of the graph be $v_1, v_2, \dots, v_{12}$ following along the main cycle, as shown in Figure 3. Assume that $v_1, v_2, \dots, v_6$ are filled. Since v_7 is the only neighbor of v_4 that is not filled, v_7 becomes filled. Then v_8 is the only neighbor of v_5 that is not filled, v_8 becomes filled. This continues until all of the vertices are filled. Hence the original set of six filled vertices is a zero forcing set of the graph.

Figure 3: A string of 6 filled vertices which forms a zero forcing set of $C_{12}\{1, 3\}$

We generalize the above example in our next lemma.

Lemma 8. A coloring of $C_n\{1, k\}$ where $n \geq 2k + 2$ with a string of $2k$ filled vertices along the main cycle forms a zero forcing set.

Proof. Assume we have $C_n\{1, k\}$ with a string of $2k$ filled vertices. Let the vertices be $v_1, v_2, v_3, \dots, v_n$ where the vertices between v_1 and v_{2k} are filled. The vertex v_{k+1} is adjacent to v_1, v_k, v_{k+2} , and v_{2k+1} . Since v_{2k+1} is the only unfilled vertex adjacent to v_{k+1} it will be forced, then v_{k+2} forces v_{2k+2} and this continues until we reach v_n and all of the vertices in the graph are filled. Hence the original set of $2k$ filled vertices is a zero forcing set of the graph. $\square$

In our previous lemma we showed that if there is a string of $2k$ vertices the graph will be forced. It can be the case that a string of $2k - 1$ vertices may not force the graph. However we show next that if there

is a string of $2k - 1$ filled vertices that is next to a string of $i \geq 2$ vertices (with an unfilled vertex in the middle) the graph is forced.

Lemma 9. Let $2 \leq i < k$. A coloring of $C_n\{1, k\}$ with a $(2k - i) - i$ configuration is a zero forcing set.

Proof. Let $v_1, v_2, \dots, v_{i-1}$ be the first string of filled vertices, let $i < k$ and v_i be the unfilled vertex separating the strings, and let $v_{i+1}, v_{i+2}, \dots, v_{2k}$ be the second string of filled vertices. Then v_{k+1} is filled and is adjacent to the filled vertices v_1, v_k , and v_{k+2} and the unfilled vertex v_{2k+1} . Then the vertex v_{2k+1} will be forced. Then in a similar fashion v_{k+t} forces v_{2k+t} for $2 \leq t \leq i$. This will create a string of $2k$ vertices along the main cycle of the graph which forms a zero forcing set. $\square$

We note that when $k \geq 3$ it is possible to have the configuration $(2k - 1) - (k - 2)$ where no forcing occurs. However in this case it must be that the string next to the string of $k - 2$ filled vertices has only a single filled vertex. As a result the configuration $(2k - 1) - (k - 2)$ will not force additional vertices. Hence they must form a configuration of the form $1 - (k - 2) - (2k - 1) - (k - 2) - 1$.

Recall a module of order X to be a set of X vertices that have the same set of neighbors in $V - X$. We next apply this property to the family of circulant graphs.

Lemma 10. If $G = C_{2(k+1)}\{1, k\}$, G then contains a module of order 2.

Proof. Suppose $G = C_{2(k+1)}\{1, k\}$. Denote the vertices along G as $v_1, v_2, \dots, v_{2(k+1)}$. Consider v_1 and v_{k+2} . v_1 is immediately adjacent to v_2 and $v_{2(k+1)}$. Since G is of the form $G = C_n\{1, k\}$, v_1 is also adjacent to $v_{2(k+1)-(k-1)} = v_{k+3}$ and v_{k+1} . v_{k+2} is immediately adjacent v_{k+1} and v_{k+3} . Again we consider its other neighbors, $v_{k+2-k} = v_2$ and $v_{k+2+k} = v_{2(k+1)}$. As v_1 and v_{k+2} share all the same neighbors, they form a module of order 2. $\square$

Theorem 11. Suppose $G = C_{2(k+1)}\{1, k\}$, then $F(G) = 2k$.

Proof. By Lemma 10, G contains a module of order 2. By Lemma 10, $F(G) = |V(G)| - 2 = 2(k + 1) - 2 = 2k$. $\square$

In Theorem 15 we give a generalization of Theorem 11. However we need first present a few lemmas.

Lemma 12. Let $G = C_n\{1, k\}$ where $2 \leq k \leq n$. If G contains a configuration $m - l$ where $m + l > 2k$, $m \geq k - 1$, $l \geq k - 1$ then the graph is forced.

Proof. Let the configuration have filled vertices $v_1, v_2, \dots, v_m$, unfilled vertex v_{m+1} , and filled vertices $v_{m+2}, v_{m+3}, \dots, v_{m+l+1}$. Then vertex v_m has three filled neighboring vertices v_{m-k} , v_{m-1} , and v_{m+k} and one unfilled vertex v_{m+1} . By the forcing rule v_{m+1} will be filled, creating a string of $m + l + 1 \geq 2k$ vertices which forces the graph by Lemma 8 $\square$

We next show for $G = C_n\{1, k\}$ if there is a configuration with three strings where the total number of vertices exceeds $3k$ and the longest of these strings is either the first or the third string the graph is forced.

Lemma 13. Let $G = C_n\{1, k\}$ where $2 \leq k \leq n$. If G contains a configuration $m - l - t$ where $m + l + t > 3k$ and $m > l$ and $m > t$, then the graph is forced.

Proof. Assume G contains a configuration $m - l - t$ where $m + l + t > 3k$ and $m > l$ and $m > t$. Then $m \geq k + 1$ and also $m \leq 2k - 1$ or the graph would be forced by Lemma 8. Then we will assume that $l \leq k - 2$, since otherwise the graph would be forced by the above lemma. Consider the string with m filled vertices where the last $k + 1$ vertices are $v_1, v_2, \dots, v_{k+1}$. Then v_{k+2} is not filled. Now if v_{2k+1} is filled, then v_{k+1} would force v_{k+2} . We can consider the cases where v_{2k+1} would not be filled. The first case involves a string of $k - 2$ vertices $v_{k+3}, v_{k+4}, \dots, v_{2k}$. Another case would involve $k - 3$ vertices that appear in the next two strings $v_{k+3}, v_{k+4}, \dots, v_{k+3+i}$ and $v_{k+4+i}, v_{k+5+i}, \dots, v_{2k}$ where $0 \leq i \leq k - 4$. $\square$

Lemma 14. Let $G = C_n\{1, k\}$ where $2 \leq k \leq n$. If G contains a configuration $m - l - t - u - v$ where $m + l + t + u + v > 5k$, then the graph is forced.

Assume G contains a configuration $m - l - t - u - v$ where $m + l + t + u + v > 5k$. Then there is some string s of length at least $k + 1$ and at most $2k - 1$. Then the sum of the vertices in the two segments to its left or right would have to be less than or equal to $k - 1$ (or the the entire graph would be forced, by the proof of Lemma 13 and Lemma 8). But then the number of vertices across these 5 segments would be at most $(2k - 1) + (k - 1) + (k - 2) + (k + 3) = 5k - 1$ which is a contradiction.

Theorem 15. Let $G = C_{N(k+1)}\{1, k\}$ where N is a positive integer. Then $F(G) = Nk$.

Proof. We first show that $F(G) \geq Nk$. Let $V(C_{N(k+1)}\{1, k\}) = \{v_1, v_2, \dots, v_n\}$. Then vertices $v_{(k+1)(i-1)+1}, v_{(k+1)(i-1)+2}, \dots, v_{(k+1)(i-1)+k}$ can be filled and vertices $v_{(k+1)i}$ are left unfilled for all $1 \leq i \leq N$. This is a failed zero forcing set since each filled vertex at the end of a string is adjacent to two unfilled vertices and all other vertices are not adjacent to any filled vertices. Next we show that $F(G) \leq Nk$. Assume that $F(G) > Nk$. Then there will be at most $N - 1$ unfilled vertices. Then we have to place $N(k + 1) - (N - 1) = Nk + 1$ filled vertices into $N - 1$ subdivisions between the unfilled vertices. Note that this means that the average number of filled vertices in a subdivision is greater than k . Then there must be a configuration where there are more than $5k$ vertices across some configuration of 5 strings. By the Lemma 14, the graph will be forced. $\square$

Theorem 16. Let $G = C_n\{1, 2\}$. Then $S \subseteq V(G)$ is a zero forcing set if and only if S either contains a string with at least four vertices or contains a configuration of the form $k - l$ where $k \geq 3$ and $l \geq 1$.

Proof. If there is string of four or more filled vertices then the graph then by Lemma 8 the graph is forced. Next, assume S contains a $k - l$ configuration where $k \geq 3$ and $l \geq 1$. If $k \geq 4$ then G is forced by Lemma 8. We proceed with $k = 3$. Let these filled vertices be v_1, v_2 , and v_3 , let v_4 be a unfilled vertex, and v_5 be a filled vertex. Then v_3 has three filled neighbors v_1, v_2 , and v_5 and one unfilled neighbor v_4 . Hence v_4 will be forced, creating a string of four consecutive filled vertices. Thus by Lemma 8 the graph will be forced. For the other direction, assume that the only strings of filled vertices in S have size 1 or 2. We will show that the graph cannot be forced. If a vertex u is a string with 1 vertex it has two unfilled neighbors. If u_i and u_{i+1} form a string of size 2. Then u_{i-1} and u_{i+2} are unfilled vertices. Then both u_i and u_{i+1} each have two unfilled neighbors. Hence S is a failed zero forcing set of G . $\square$

In Figure 4 we show failed zero forcing sets for $C_{n-1}\{1, 2\}$ for $n = 5, 6$ and 7 .

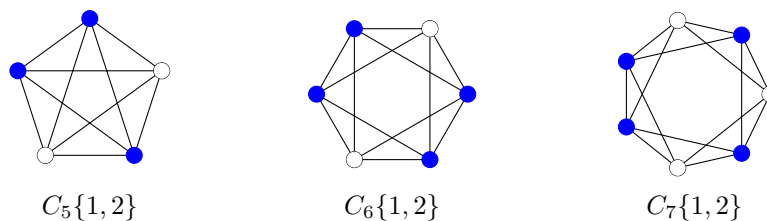
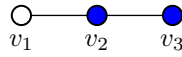


Figure 4: Failed zero forcing sets for small values of n

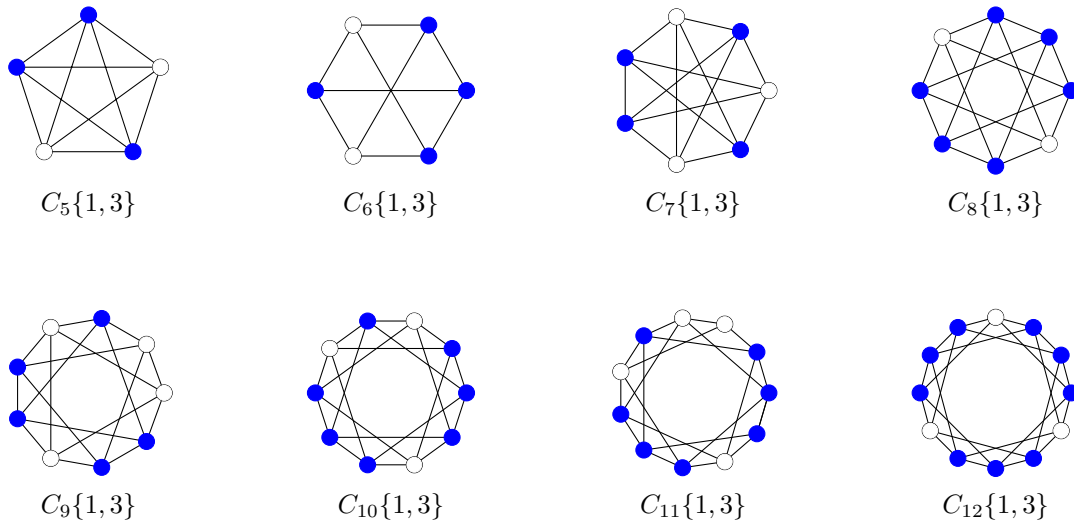
Theorem 17. If $G = C_n\{1, 2\}$ where $n \geq 5$, $F(C_{3t}\{1, 2\}) = 2t$, $F(C_{3t+1}\{1, 2\}) = 2t$, $F(C_{3t+2}\{1, 2\}) = 2t + 1$.

Figure 5: A string that can be used to extend a failed zero forcing set of $C_n(1, 2)$

Proof. We explicitly state the failed zero forcing sets for each case. Let the vertices in C_n be $v_1, v_2, \dots, v_n$. When $n = 3t$ a failed zero forcing with $2t$ vertices can be formed with vertices $v_1, v_2, v_4, v_5, \dots, v_{3i-2}, v_{3i-1}, \dots, v_{3t-2}, v_{3t-1}$. When $n = 3t + 1$ a failed zero forcing set with $2t$ vertices can be formed with vertices $v_1, v_2, v_4, v_5, \dots, v_{3i-2}, v_{3i-1}, \dots, v_{3t-2}, v_{3t-1}$. When $n = 3t + 2$ a failed zero forcing set with $2t + 1$ vertices can be formed with vertices $v_1, v_2, v_4, v_5, \dots, v_{3i-2}, v_{3i-1}, \dots, v_{3t-2}, v_{3t-1}$.

Next we show the upper bound. Note that if there was a failed zero forcing set with three consecutively filled vertices, then there must be two adjacent unfilled vertices at each end. This would be a total of seven vertices which could be replaced by a $2 - 2$ configuration with an unfilled vertex at each end. So to have a maximal failed zero forcing set it must be the case that at most two out of every three consecutive vertices must be filled. This creates an upper bound of $\lfloor \frac{2n}{3} \rfloor$. Hence $F(C_{3t}\{1, 3\}) \leq \lfloor \frac{2(3t)}{3} \rfloor = 2t$, $F(C_{3t+1}\{1, 3\}) \leq \lfloor \frac{2(3t+1)}{3} \rfloor = 2t$, and $F(C_{3t+2}\{1, 3\}) \leq \lfloor \frac{2(3t+2)}{3} \rfloor = 2t + 1$. □

Next we consider $F(C_n\{1, 3\})$. We note that $C_4\{1, 3\} = K_4$, so $F(C_4\{1, 3\}) = F(K_4) = 2$. In Figure 6, we give lower bounds for $C_n\{1, 3\}$ for $5 \leq n \leq 12$.

Figure 6: Maximum failed zero forcing sets for $C_n\{1, 3\}$ for small values of n

In our next lemma we consider conditions for inserting a $3 - 3$ configuration with an adjacent unfilled vertex (shown in Figure 2) into $C_n\{1, 3\}$ to create a larger failed zero forcing set.

Lemma 18. Let $n \geq 5$ and let S be a failed zero forcing set of $C_n\{1, 3\}$ of size k with three consecutive vertices v_1, v_2 and v_3 along the main cycle where v_1 and v_3 are filled and v_2 is unfilled. Furthermore assume

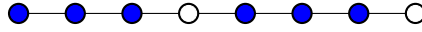


Figure 7: A 3-3 path

that either v_n or v_{n-2} is unfilled and assume that either v_n or v_2 is unfilled. Then a $3-3$ string can be inserted between v_2 and v_3 , resulting in a failed zero forcing set of size $k+6$.

Proof. Let the $3-3$ configuration with an additional unfilled vertex on one end of consecutive vertices include $u_1, u_2, u_3, u_4, u_5, u_6, u_7$, and u_8 , where all of these vertices are filled except for u_4 and u_8 . We verify that the extended set is a failed zero forcing set by showing that each of the filled vertices either have zero or at least two unfilled neighbors. The vertex v_1 is adjacent to v_l and v_2 ; u_1 is adjacent to v_2 and u_4 ; u_3 has no unfilled neighbors; u_5 is adjacent to u_4 and u_8 , u_6 has no unfilled neighbors; and u_7 is adjacent to u_4 and u_8 , and v_3 is adjacent to u_8 and v_r . $\square$

Lemma 19. If $G = C_{n+8i}(1, 3)$ where $n \geq 5$ and $i \geq 0$ we obtain the following lower bounds:

$$\begin{aligned} F(C_{5+8i}\{1, 3\}) &\geq 3 + 6i, \\ F(C_{6+8i}\{1, 3\}) &\geq 4 + 6i, \\ F(C_{7+8i}\{1, 3\}) &\geq 4 + 6i, \\ F(C_{8+8i}\{1, 3\}) &\geq 6 + 6i, \\ F(C_{9+8i}\{1, 3\}) &\geq 5 + 6i, \\ F(C_{10+8i}\{1, 3\}) &\geq 7 + 6i, \\ F(C_{11+8i}\{1, 3\}) &\geq 7 + 6i, \text{ and} \\ F(C_{12+8i}\{1, 3\}) &\geq 9 + 6i. \end{aligned}$$

Proof. The proof follows by repeated applications of Lemma 20. $\square$

In the next three lemmas, we give configurations that will force the circulant graph $C_n\{1, 3\}$.

Lemma 20. In $C_n\{1, 3\}$, a $5-l$ configuration with $l \geq 2$ will force the graph.

Proof. Assume that on a graph $C_n\{1, 3\}$, there exists a $5-l$ configuration with $l \geq 2$. The first five vertices can be named v_1, v_2, v_3, v_4 and v_5 . We can call the unfilled vertex v_6 . Let the next two filled vertices be v_7 and v_8 . Since v_5 has exactly three filled neighbors v_2, v_4 , and v_8 the unfilled neighbor v_6 will be forced. We now have a path of 7 filled vertices which forces the graph by Lemma 19. We note this result also holds when the second path consists of more than two consecutive filled vertices. $\square$

Lemma 21. On $C_n\{1, 3\}$, a $4-l$ configuration with $l \geq 2$ will force the graph.

Proof. Assume that $C_n\{1, 3\}$, contains a configuration of the form $4-l$ where $l \geq 2$. Let the vertices in this configuration be $v_1, v_2, v_3, v_4, v_5, v_6$ and v_7 . Then v_5 will be the only unfilled neighbor of v_4 , and so it will be filled by the forcing rule. Then we have a path of 7 filled vertices which forces the graph by Lemma 8. $\square$

Lemma 22. Let $C_n\{1, 3\}$ contain a $3-2$ configuration. along the main cycle adjacent on the right to a path of at least one filled vertex also along the main cycle forces the graph. Similarly, a $2-3$ configuration along the main cycle adjacent on the left to a path of at least one filled vertex also along the main cycle forces the graph.

Proof. Let v_1, v_2, v_3, v_5 , and v_6 denote a set of 5 filled vertices. Let v_4 denote the unfilled vertex separating the filled in vertices, and v_7 denote the remaining unfilled vertex that is adjacent to v_6 . Suppose we have a path of at least one filled vertex on the right immediately adjacent v_7 . Now v_5 is adjacent to the

first filled vertex in this new path. v_5 is also adjacent to v_2 , and the unfilled vertex v_4 . As v_4 is the only unfilled neighbor of v_5 , v_4 is forced and v_1, v_2, v_3, v_4, v_5 , and v_6 form a path of 6 filled vertices along the main cycle which forces the graph by Lemma 8. The case with a $2-3$ configuration follows by a symmetric argument. $\square$

Lemma 23. Let $G = C_{n+8i}\{1, 3\}$ where $i \geq 0$ and $5 \leq n \leq 12$. Then we have the following upper bounds:

$$\begin{aligned} F(C_{5+8i}\{1, 3\}) &\leq 3 + 6i, \\ F(C_{6+8i}\{1, 3\}) &\leq 4 + 6i, \\ F(C_{7+8i}\{1, 3\}) &\leq 4 + 6i, \\ F(C_{8+8i}\{1, 3\}) &\leq 6 + 6i, \\ F(C_{9+8i}\{1, 3\}) &\leq 5 + 6i, \\ F(C_{10+8i}\{1, 3\}) &\leq 7 + 6i, \\ F(C_{11+8i}\{1, 3\}) &\leq 7 + 6i, \text{ and} \\ F(C_{12+8i}\{1, 3\}) &\leq 9 + 6i. \end{aligned}$$

Proof. We verify each of these cases individually, making repeated use of the pigeonhole principle.

(i) $F(C_{5+8i}\{1, 3\}) \leq 3 + 6i$ for all $i \geq 0$. Assume that $F(C_{5+8i}) \geq 4 + 6i$. Then we have to distribute $4 + 6i$ filled vertices over $1 + 2i$ subdivisions. We will first assume that none of the subdivisions contain 0 vertices. Repeatedly removing strings with 6 vertices distributed over 2 adjacent subdivisions results in 4 vertices to be distributed over 1 subdivision, unless we have a $1-4-1$ configuration, the graph will be forced. If we do have we remove this from the original labeling. This would leave us to distribute $6i - 2$ vertices over $2i - 2$ subdivisions. Then removing $i - 1$ pairs (each totaling 6 filled vertices) leaves 4 vertices to be distributed over 0 subdivisions, which is impossible. If it was the case that there was a subdivision with 0 vertices then distributing $4 + 6i$ filled vertices over $1 + 2i$ subdivisions would leave 4 vertices to be distributed over 0 subdivisions, which again is impossible.

(ii) $F(C_{6+8i}\{1, 3\}) \leq 4 + 6i$ for all $i \geq 0$. Assume that $F(C_{6+8i}) \geq 5 + 6i$. Then we have to distribute $5 + 6i$ filled vertices over $1 + 2i$ subdivisions. Repeatedly removing strings with 6 vertices distributed over 2 adjacent subdivisions results in 4 vertices to be distributed over 1 subdivision, which will force the graph.

(iii) $F(C_{7+8i}\{1, 3\}) \leq 4 + 6i$ for all $i \geq 0$. Assume that $F(C_{7+8i}) \geq 5 + 6i$. Then we have to distribute $5 + 6i$ filled vertices over $2 + 2i$ subdivisions. We will assume that none of the subdivisions contain 0 vertices. Repeatedly removing strings with 6 vertices distributed over 2 adjacent subdivisions results in 5 vertices to be distributed over 2 subdivisions, which will force the graph.

(iv) $F(C_{8+8i}\{1, 3\}) \leq 6 + 6i$ for all $i \geq 0$. Assume that $F(C_{8+8i}) \geq 7 + 6i$. Then we have to distribute $7 + 6i$ filled vertices over $1 + 2i$ subdivisions. We will assume that none of the subdivisions contain 0 vertices. Repeatedly removing strings with 6 vertices distributed over 2 adjacent subdivisions results in 7 vertices to be distributed over 1 subdivision, which will force the graph.

(v) $F(C_{9+8i}\{1, 3\}) \leq 5$. Assume $F(C_9) \geq 6$. Then we have at least 6 filled vertices distributed over 3 subdivisions. Since there are 9 vertices in the graph there would have to be exactly 6 filled vertices distributed over 3 subdivisions. Any permutations of the adjacent configurations $2-1-3$, $2-2-2$, $4-1-1$, $4-0-2$, or $5-1-0$ will force the graph. This yields a contradiction. We next show that $F(C_{9+8i}) \leq 5 + 6i$ for all $i \geq 1$. Assume $F(C_{9+8i}) \geq 6 + 6i$. Then we have at least $6 + 6i$ filled vertices distributed over $3 + 2i$ subdivisions. Repeatedly subtracting strings with 6 vertices distributed over 2 adjacent subdivisions results in 6 filled vertices distributed over 3 subdivisions. Any permutations of the configurations $2-2-2$, $4-1-1$, $4-0-2$, or $5-1-0$ will force the graph. However we must consider the case of $2-1-3$. This would have to be part of one of the configurations $0-2-1-3$, $1-2-1-3$, $2-2-1-3$, $3-2-1-3$, $4-2-1-3$, or $5-2-1-3$. All but the first two cases force the graph. If the original labeling contained configurations of the form $0-2-1-3$ or $1-2-1-3$. Then we could start by removing the configurations $0-2-1$ or $1-2-1$ and then strings with 6 vertices distributed over 2 adjacent subdivisions. This would leave at least 2 filled vertices distributed over zero subdivisions which yields a contradiction.

(vi) $F(C_{10+8i}\{1,3\}) \leq 7+6i$ for all $i \geq 0$. Assume that $F(C_{10+8i}) \geq 8+6i$. Then we have to distribute $8+6i$ filled vertices over $2+2i$ subdivisions. We will assume that none of the subdivisions contain 0 vertices. Repeatedly removing strings with 6 vertices distributed over 2 adjacent subdivisions results in 8 vertices to be distributed over 2 subdivisions, which will force the graph.

(vii) $F(C_{11+8i}\{1,3\}) \leq 7+6i$ for all $i \geq 0$. Assume that $F(C_{11+8i}) \geq 8+6i$. Then we have to distribute $8+6i$ filled vertices over $3+2i$ subdivisions. We will assume that none of the subdivisions contain 0 vertices. Repeatedly removing strings with 6 vertices distributed over 2 adjacent subdivisions results in at least 8 vertices to be distributed over 3 subdivisions, which will force the graph. We consider different subcases.

(a) If there are 11 vertices distributed over 3 subdivisions it be a configuration of form $5-1-5$. But then this is part of a $5-1-5-1$ or $1-5-1-5$ configuration. Then if we removed a $1-5-1$ configuration from the original coloring, we would be left to distribute $1+6i$ over $2i$ subdivisions, which after repeatedly removing strings with 6 vertices distributed over 2 adjacent subdivisions results in 1 vertex over 0 subdivisions, which is impossible.

(b) If there are 10 vertices distributed over 3 subdivisions, we must either have a 4 or a 5 adjacent to a number larger than 1. Either of these would force the graph.

(c) If there are 9 vertices distributed over 3 subdivisions, the only configuration that would not force the graph is $3-3-3$. Since n is not a multiple of 4, there would have to be a 3 next to a 1, 2, 4, or 5. The last three would force the graph. If we had a $3-1$ configuration we could remove it from the original graph which would leave $4+6i$ vertices to distribute over $2i+1$ subdivisions. After repeatedly removing strings with 6 vertices distributed over 2 adjacent subdivisions results in 4 filled vertices over 1 subdivision which would force the graph.

(d) If there are 8 vertices distributed over 3 subdivisions, the only case that would not force the graph is the configuration $5-2$. The only cases that would not force the graph is if the configuration $5-2$ was part of a $5-2-1$ configuration or a $5-2-2$ configuration. In these cases we could remove the following parts of the configuration $1-2-1$ or $1-2-2$. This would leave at least $3+6i$ filled vertices to distribute over $2i$ subdivisions. After repeatedly removing strings with 6 vertices distributed over 2 adjacent subdivisions results in 3 filled vertices over 0 subdivisions, which is impossible.

(viii) $F(C_{12+8i}) \leq 9+6i$ for all $i \geq 0$. Assume that $F(C_{12+8i}) \geq 10+6i$. Then we have to distribute $10+6i$ filled vertices over $2+2i$ subdivisions. We will assume that none of the subdivisions contain 0 vertices. Repeatedly removing strings with 6 vertices distributed over 2 adjacent subdivisions results in 10 vertices to be distributed over 2 subdivisions, which will force the graph. $\square$

We can combine Lemmas 19 and 23 to obtain the following result.

Theorem 24. *Let $G = C_{n+8i}\{1,3\}$ where $i \geq 0$ and $5 \leq n \leq 12$. Then we have the following upper bounds:*

$$F(C_{5+8i}\{1,3\}) = 3+6i$$

$$F(C_{6+8i}\{1,3\}) = 4+6i$$

$$F(C_{7+8i}\{1,3\}) = 4+6i$$

$$F(C_{8+8i}\{1,3\}) = 6+6i$$

$$F(C_{9+8i}\{1,3\}) = 5+6i$$

$$F(C_{10+8i}\{1,3\}) = 7+6i$$

$$F(C_{11+8i}\{1,3\}) = 7+6i, \text{ and}$$

$$F(C_{12+8i}\{1,3\}) = 9+6i.$$

Next we investigate failed zero forcing numbers for $C_n\{1,2,3\}$. We begin with two lemmas.

Lemma 25. If $C_n\{1,2,3\}$ has a coloring with a $k-l$ configuration where $k+l=6$, $k \neq l$, then the graph is forced.

Proof. Assume that there are $k \geq 6$ filled vertices $v_1, v_2, \dots, v_k$ that appear consecutively and there is an unfilled vertex v_{k+1} adjacent to v_k . Then the vertex v_{k-2} is adjacent to five filled vertices $v_{k-5}, v_{k-4}, v_{k-3}, v_{k-1}$, and v_k and only one unfilled vertex v_{k+1} . Hence the unfilled vertex will be forced. Since every set of k filled vertices can be extended all of the vertices in the graph will be forced. $\square$

Lemma 26. If $C_n\{1, 2, 3\}$ has 6 or more filled vertices distributed into two adjacent strings of different sizes over two subdivisions then the graph is forced.

Proof. We consider two cases with the configurations, $4 - 2$ and $5 - 1$. Case 1: Assume that there are four filled vertices v_1, v_2, v_3 , and v_4 in the first string followed by an unfilled vertex v_5 , followed by two filled vertices v_6 and v_7 . Then v_4 is adjacent to five filled vertices v_1, v_2, v_3, v_6 , and v_7 and one unfilled vertex v_5 . Hence v_5 will be forced to be filled creating a string of 6 consecutive filled vertices which will force the graph.

Case 2: Assume that there are five filled vertices v_1, v_2, v_3, v_4 and v_5 in the first string followed by an unfilled vertex v_6 , followed by a filled vertex v_7 . Then v_4 is adjacent to five filled vertices v_1, v_2, v_3, v_5 , and v_7 and one unfilled vertex v_6 . Hence v_6 will be forced to be filled creating a string of 6 consecutive filled vertices which will force the graph. $\square$

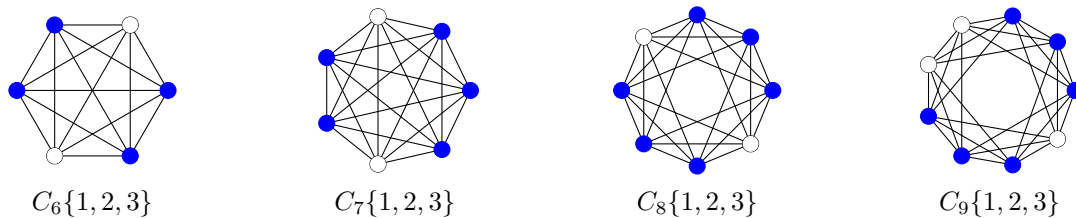


Figure 8: $F(C_n\{1, 2, 3\})$ for small values of n

The next two lemmas give lower and upper bounds for certain 6-regular circulant graphs.

Lemma 27. If $G = C_{n+4i}\{1, 2, 3\}$, where $n \geq 6$ and $i \geq 0$ then we obtain the following upper bounds:

$$\begin{aligned} F(C_{6+4i}\{1, 2, 3\}) &\geq 4 + 3i \\ F(C_{7+4i}\{1, 2, 3\}) &\geq 5 + 3i, \\ F(C_{8+4i}\{1, 2, 3\}) &\geq 6 + 3i, \text{ and} \\ F(C_{9+4i}\{1, 2, 3\}) &\geq 6 + 3i. \end{aligned}$$

Proof. The first four cases are shown in Figure 8. All of these cases can be extended repeatedly by inserting a $3 - 1$ string in between a unfilled vertex and a filled vertex that is located next to it in a counterclockwise direction. $\square$

Lemma 28. If $G = C_{n+4i}\{1, 2, 3\}$, where $n \geq 6$ and $i \geq 0$ then we obtain the following upper bounds:

$$\begin{aligned} F(C_{6+4i}\{1, 2, 3\}) &\leq 4 + 3i \\ F(C_{7+4i}\{1, 2, 3\}) &\leq 5 + 3i, \\ F(C_{8+4i}\{1, 2, 3\}) &\leq 6 + 3i, \text{ and} \\ F(C_{9+4i}\{1, 2, 3\}) &\leq 6 + 3i. \end{aligned}$$

Proof. If any of these conditions fail to hold then we will have have: $5 + 3i$ filled vertices to be distributed over $1 + i$ subdivisions; $6 + 3i$ filled vertices to be distributed over $1 + i$ subdivisions; $7 + 3i$ filled vertices to be distributed over $1 + i$ subdivisions; or $7 + 3i$ filled vertices to be distributed over $2 + i$ subdivisions. In all

of these cases the average number of filled vertices per subdivision will be greater than 3. Hence there will be two adjacent strings with more than 6 vertices. Hence, by the Lemma 26 a the graph will be forced. $\square$

The combination of Lemmas 27 and 28 yields the following result.

Theorem 29. *If $G = C_{n+4i}\{1, 2, 3\}$, where $n \geq 6$ and $i \geq 0$ then we obtain the following upper bounds:*

$$F(C_{6+4i}\{1, 2, 3\}) = 4 + 3i$$

$$F(C_{7+4i}\{1, 2, 3\}) = 5 + 3i,$$

$$F(C_{8+4i}\{1, 2, 3\}) = 6 + 3i, \text{ and}$$

$$F(C_{9+4i}\{1, 2, 3\}) = 6 + 3i.$$

Above we have investigated circulant graphs that are 4-regular. Next we investigate zero forcing sets and failed zero forcing sets for a broader class of circulant graphs, graphs with the form $C_n\{1, 2, \dots, k\}$, where $n \geq 2k$.

Our next lemma is a generalization of Lemma 8.

Lemma 30. Let $G = C_n\{1, 2, \dots, k\}$, where $n \geq 2k$. If a subset S of $V(G)$ contains $2k$ consecutive filled vertices on the main cycle, then S is a zero forcing set of G .

Proof. Let the vertices of G be $v_1, v_2, \dots, v_n$ labeled as they appear on the main cycle. Let $S = v_i, v_{i+1}, \dots, v_{i+2k-1}$ be a subset of filled vertices. Then v_{i+2k} is the only neighbor of v_{i+k} that is not filled, so it becomes forced. Then in a similar fashion $v_{i+2k+1}, v_{i+2k+2}, \dots, v_n$ sequentially are forced. $\square$

Lemma 31. The graph $C_n\{1, 2, \dots, k\}$, is not forced by a $k - k$ configuration, where $k \leq \lfloor \frac{n}{2} \rfloor - 1$.

Proof. Assume $v_1, v_2, \dots, v_k$ are all filled and v_n and v_{k+1} are unfilled. Suppose we have an adjacent string with k vertices $v_{k+2}, v_{k+3}, \dots, v_{2k+1}$ filled and v_{2k+2} is unfilled. Since each vertex is adjacent to every vertex with a distance of $1, 2, \dots, k$, then every filled vertex will be adjacent to two unfilled vertices v_n, v_{k+1} , or v_{2k+2} . Thus the graph is not forced by a $k - k$ configuration. $\square$

Lemma 32. In $C_n\{1, 2, \dots, k\}$ for all $k \leq \lfloor \frac{n}{2} \rfloor - 1$ the configuration $k - k$ is not a failed zero forcing set.

Proof. Let $V(C_n\{1, 2, \dots, k\}) = \{v_1, v_2, \dots, v_n\}$. Assume we have $C_n\{1, 2, \dots, k\}$ with a path of k filled vertices. Let these vertices be $v_1, v_2, \dots, v_k$. Then v_n and v_{k+1} are left unfilled. Suppose we have a path of $k + 1$ filled vertices next to this k path. Let these vertices be $v_{k+2}, v_{k+3}, \dots, v_{2k+3}$. Then v_{2k+4} is unfilled. Notice each vertex will be adjacent to all vertices with distances of $1, 2, \dots, k$. This means vertex v_{k+2} will be adjacent to vertices $\{v_{k+3}, v_{k+4}, \dots, v_{2k+2}\}$ and $\{v_2, v_3, \dots, v_k, v_{k+1}\}$. Since v_{k+2} has only one unfilled neighbor, v_{k+1} , then this vertex will be forced. Since v_{k+1} is now filled, then v_k has only one unfilled neighbor, v_n . This forces v_n , and so we have a row of $2k + 4$ filled vertices. We can continue with v_{k+4} forcing v_{2k+4} and continuing until every vertex in G is forced. Thus we cannot have a $k - k$ configuration without forcing the graph. $\square$

Next we give a lower bound for a generalized family of circulant graphs.

Theorem 33. *If $G = C_n\{1, 2, \dots, k\}$, then $F(G) \geq \lfloor \frac{kn}{k+1} \rfloor$, where $4 \leq k \leq \lfloor \frac{n}{2} \rfloor - 1$.*

Proof. Let the vertices be $v_1, v_2, \dots, v_n$. If $n = t(k + 1) + i$ where $0 \leq i \leq k$ then we can use a labeling where vertices $v_{j(k+1)}$ for all $1 \leq j \leq \lfloor \frac{n}{k+1} \rfloor$ and v_n are unfilled and the remaining vertices are filled. The number of unfilled vertices is t if $(k + 1) | n$ and $t + 1$ if $(k + 1) \nmid n$. If $(k + 1) | n$ then the number of filled vertices is $n - \frac{n}{k+1} = \frac{n(k+1)}{k+1} - \frac{n}{k+1} = \frac{nk}{k+1}$. If $(k + 1) \nmid n$ then we assume that v_n is unfilled to avoid a $k - l$ configuration

where $l \geq k+1$. If $(k+1) \nmid n$ then the number of filled vertices is $n - \left(\left\lfloor \frac{n}{k+1} \right\rfloor + 1\right) = \frac{n(k+1)}{k+1} - \left\lfloor \frac{n}{k+1} \right\rfloor - 1 = \left\lfloor \frac{nk}{k+1} \right\rfloor$.

To see that the filled vertices form a failed zero forcing set, note that each filled vertex has at least two unfilled neighbors. $\square$

Next, we present an upper bound for a generalized family of circulant graphs.

Theorem 34. *If $G = C_n\{1, 2, \dots, k\}$, then $F(C_{n+i(k+1)}\{1, 2, \dots, k\}) \leq F(G) + ki$, where $4 \leq k \leq \lfloor \frac{n}{2} \rfloor - 1$ and $n \geq 10$.*

Proof. Suppose $F(C_{n+i(k+1)}\{1, 2, \dots, k\}) = F(G) + ki + 1$. By Theorem 33, we know $F(G) \geq \lfloor \frac{kn}{k+1} \rfloor$, then there are $(n + i(k+1)) - (\lfloor \frac{kn}{k+1} \rfloor + ki + 1) = n + i - 1 - \lfloor \frac{kn}{k+1} \rfloor \notin S$. Again, we will refer to the set of vertices on the main cycle between unfilled vertices as subdivisions, where each subdivision can have a finite number of filled vertices. We must then distribute $\lfloor \frac{kn}{k+1} \rfloor + k(i) + 1$ filled vertices over $n + i - 1 - \lfloor \frac{kn}{k+1} \rfloor$ subdivisions. Using Lemma 32, we can have a string of k consecutively filled vertices within a subdivision without forcing the graph, so we can distribute $k(i)$ filled vertices across i subdivisions. We are left with $\lfloor \frac{kn}{k+1} \rfloor + 1$ filled vertices over $n - 1 - \lfloor \frac{kn}{k+1} \rfloor$ subdivisions.

Recall $k \leq \lfloor \frac{n}{2} \rfloor - 1$. There are two cases, (i) when n is even and (ii) when n is odd.

(i) Suppose n is even, then $k = \frac{n}{2} - 1$. So, we have $n - 1 - \lfloor \frac{kn}{k+1} \rfloor = n - 1 - \lfloor \frac{(\frac{n}{2}-1)n}{(\frac{n}{2}-1)+1} \rfloor = n - 1 - \lfloor \frac{(\frac{n}{2}-1)n}{\frac{n}{2}} \rfloor = n - 1 - \lfloor 2(\frac{n}{2} - 1) \rfloor = n - 1 - n + 2 = 1$ subdivision left to fit $\lfloor \frac{kn}{k+1} \rfloor + 1 = 2(\frac{n}{2} - 1) + 1 = n - 1 = 2k$ filled vertices. Since $k < \lfloor \frac{kn}{k+1} \rfloor + 1$, then there a set of $2k$ consecutively filled vertices. By Lemma 8, this forces the graph which is a contradiction.

(ii) Suppose n is odd. Let $n = 2t + 1$, where $t \in \mathbb{N}$. Then $k = \lfloor \frac{n}{2} \rfloor - 1 = \lfloor \frac{2t+1}{2} \rfloor - 1 = t - 1 = \frac{n-1}{2} - 1 = \frac{n-3}{2}$. Hence, we have $n - 1 - \lfloor \frac{kn}{k+1} \rfloor = n - 1 - \lfloor \frac{(\frac{n-3}{2})n}{\frac{n-3}{2}+1} \rfloor = n - 1 - \lfloor \frac{(\frac{n-3}{2})n}{\frac{n-1}{2}} \rfloor = n - 1 - \lfloor \frac{n(n-3)}{n-1} \rfloor = n - 1 - \lfloor \frac{n}{n-1}(n-3) \rfloor = n - 1 - (n-3) = 2$ subdivisions leftover. Then there are $\lfloor \frac{kn}{k+1} \rfloor + 1 = (n-3) + 1 = n-2$ filled vertices left. Which means on average each subdivision will have k filled vertices. Hence, the total sum of all filled vertices is $ki + 2k = ki + 2(\frac{n-3}{2}) = ki + n - 3$. Since this is less than $F(G) + ki + 1 = \lfloor \frac{kn}{k+1} \rfloor + ki + 1 = (n-3) + ki + 1 = ki + n - 2$, we arrive at a contradiction. Thus, $F(C_{n+i(k+1)}\{1, 2, \dots, k\}) = F(G) + ki + 1$ is impossible. $\square$

4 Conclusion

Failed zero forcing numbers are still unknown for many families of graphs. The problem of finding $F(T)$ where T is a tree (originally posed in [6]) is still open. In Theorem 2 we determined $F(G)$ where G is a subdivided star. It is likely that techniques used in the proof of Theorem 2 can be extended beyond subdivided stars. For example, failed zero forcing numbers for trees with n vertices and $n - t$ leaves can easily be obtained when t is small.

We showed in Theorem 5 that for a tree T , $F(T) \geq \alpha(T) - 1$. It would be interesting to determine if the same property could be extended to connected graphs in general. We present this as a conjecture.

Conjecture 1. $F(G) \geq \alpha(G) - 1$.

In Section 3, we obtained specific results for $F(C_n\{1, k\})$ and general results when $k = 2$ and $k = 3$. It would be interesting to determine the case where $k = 4$ and then attempt the general problem stated below.

Problem 1. Determine $F(C_n\{1, k\})$.

A problem with greater difficulty would be to determine failed zero forcing numbers for circulant graphs that are r -regular where $r \geq 4$. Another area for future study is to determine failed zero forcing numbers for Cayley graphs.

Acknowledgements

We are immensely grateful to an anonymous referee for their suggestions (including a shorter proof for Theorem 5), which helped improve the presentation of our paper. This research was done as part of the National Science Foundation Research Experiences for Undergraduates Program in Extremal Graph Theory and Dynamical Systems (Award No. 1950189) at the Rochester Institute of Technology.

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